Lecture 5 2025-09-11

Example: Let's pretend we have a 6 bit precision system

$s = 1$ sign bit

$c = 3$ exponent $\quad c_0 = 3$

$f = 2$ fraction

$$x = \begin{cases} (-1)^s 2^{c-3}(1+f) & 1 \le c \le 6 \\ (-1)^s 2^{-2} f & c = 0 \\ \pm\infty & c = 7,\ f = 0 \\ \text{NaN} & c = 7,\ f \neq 0 \end{cases}$$

Some numbers

(i) $x = 0\ 101\ 01$

$s = 0,\ c = 1\times 2^2 + 0 * 2^1 + 1 * 2^0 = 5$

$f = 0 * 2^{-1} + 1 * 2^{-2} = \frac{1}{4}$

$\Rightarrow x = (-1)^0\ 2^{5-3}\ 1.25 = 2^2 \cdot 1.25 = 5$ (in decimal)

(ii) $x = 0\ 011\ 00$

$s = 0 \qquad c = 0 * 2^2 + 1 * 2^1 + 1 \times 2^0 = 3$

$f = 0$

$x = (-1)^0\ 2^{3-3}\ 1.00 = 1$

(iii) $x = 1\ 010\ 10$

$s = 1 \qquad c = 0 * 2^2 + 1 * 2^1 + 0 * 2^0 = 2$

$f = 1 * 2^{-1} + 0 * 2^{-2} = 0.5$

$x = (-1)^1\ 2^{2-3}\ (1+0.5) = -1 \cdot \frac{1}{2} \cdot 1.5 = -0.75$

When two floating point numbers are combined (+ - * %)
the result may not be machine-representable.
IEEE guarantees correct rounding and "rounds to even"
(last digit is zero) in case of ties.

Example of rounding in 6 bit precision system:

0 101 01 ⊕ 0 010 00

floating point addition

$(-1)^0\ 2^{5-3} \cdot 1.25$ $\quad$ $(-1)^0\ 2^{2-3} \cdot 1$

5 $\quad$ 0.5 $\quad = 5.5$

$4 = 2^2 \cdot 1.00 \mathrel{\hat{=}} 0\ 101\ 00$ ← last digit is odd

$5 = 2^2 \cdot 1.25 \mathrel{\hat{=}} 0\ 101\ 01$

$6 = 2^2 \cdot 1.5 \mathrel{\hat{=}} 0\ 101\ 10$ ⇐ "even" 0 in last digit

$7 = 2^2 \cdot 1.75$

$8 = 2^3 \cdot 1.00$

⇒ 5.5 is not representable with 6 digits

⇒ We round to "even" which yields $6 \mathrel{\hat{=}} 010110$

"Round to even" means last digit is zero (only in case of a tie)

E.g. If for some reason we ended up with 4.5, we would have rounded "down" to 4 since 4 has an even last digit and 5 has an odd last digit and they are the same distance away

(adding 0 101 00 and 0 010 00)

- Numbers that have magnitude less than $2^{-1022}\cdot(1+0)$ in the 64 bit system (lowest representable number) result in underflow and are set to zero in general.

- Numbers greater than $2^{1023}\cdot(2-2^{-52})$ result in overflow and typically cause the computations to stop.

- 2 representations for 0: $s=0, c=0, f=0$
  $s=1, c=0, f=0$

Def 1.15 Suppose p^* is an approximation of p.

The actual error is $p-p^*$

- absolute error is $|p-p^*|$
- relative error is $\frac{|p-p^*|}{|p|}$ (provided $p\neq 0$)

Def 1.16 The number p^* is said to approximate p to t significant digits if t is the largest nonnegative integer for which

$$\frac{|p-p^*|}{|p|} \leq 5\cdot 10^{-t}$$

Examples: $p=\pi=3.14159265$

Show that $p^* = \begin{cases} 3.1415 & \text{approximates } \pi \text{ up to 5 significant digits} \\ 3.1416 & \text{approximates } \pi \text{ up to 6 significant digits.} \end{cases}$

$$\frac{|\pi-3.1415|}{|\pi|} = 2.9\cdot 10^{-5} \leq 5\cdot 10^{-5}$$

π

$$\frac{|\pi - 3.1416|}{\pi} = 2.34 \cdot 10^{-6} \leq 5 \cdot 10^{-6}$$

$p = 1 \quad p^* = 0.9996$

$$\frac{|p - p^*|}{|p|} = 0.0004 = 4 * 10^{-4} \leq 5 * 10^{-4}$$

$\Rightarrow$ p^* approximates p up to 4 significant digits

Main thing to watch out for: Subtracting two numbers of same sign that are nearly equal

$\Rightarrow$ Cancellation can occur.

Example: Find the roots of $x^2 - 56x + 2 = 0$

Solution:

$$x = \frac{56 \pm \sqrt{56^2 - 8}}{2} = 28 \pm \sqrt{782} = x_\pm$$

$\sqrt{782} \approx 27.964$ (pretend we have only 3 decimal digits)

$\Rightarrow x_+ \approx 55.964$

$x_- \approx 0.036$

$$\frac{|x_+ - 55.964|}{x_+} = 4.698 \cdot 10^{-6}$$

$\Rightarrow$ 55.964 approximates x_+ up to 6 significant digits.

$$\frac{|x_- - 0.036|}{x_-} = 7.36 \cdot 10^{-3}$$

$\Rightarrow$ 0.036 approximates x_- up to 2 significant digits

Sometimes cancellation can be avoided by choosing the order of floating point operations in a clever way.

$$X = \frac{-b \pm \sqrt{b^2-4ac}}{2a} \cdot \frac{-b \mp \sqrt{b^2-4ac}}{-b \mp \sqrt{b^2-4ac}} = \frac{b^2-(b^2-4ac)}{2a(-b \mp \sqrt{b^2-4ac})}$$

$$= \frac{2c}{-b \mp \sqrt{b^2-4ac}}$$

Cancellation occurs when • adding terms of opposite sign
• subtracting terms of same sign

Depending on the sign of b, we choose one or the other expression of the quadratic formula. (to avoid using extra precision in intermediate calculations.

• If $b \geq 0$ then $x_+ = \frac{2c}{-b-\sqrt{b^2-4ac}}$, $x_- = \frac{-b-\sqrt{b^2-4ac}}{2a}$

• If $b < 0$ then $x_+ = \frac{-b+\sqrt{b^2-4ac}}{2a}$, $x_- = \frac{2c}{-b+\sqrt{b^2-4ac}}$

In the previous example $b = -56 < 0$ so we should compute

$$x_- = \frac{2 \cdot 2}{56+\sqrt{56^2-8}} = \frac{2}{28+\sqrt{782}} = 0.03573726$$

($\sqrt{782} \approx 27.964$)

$$\frac{|x_- - 0.0357371|}{x_-} = 4.707 \cdot 10^{-6}$$

$\Rightarrow$ 6 significant digits